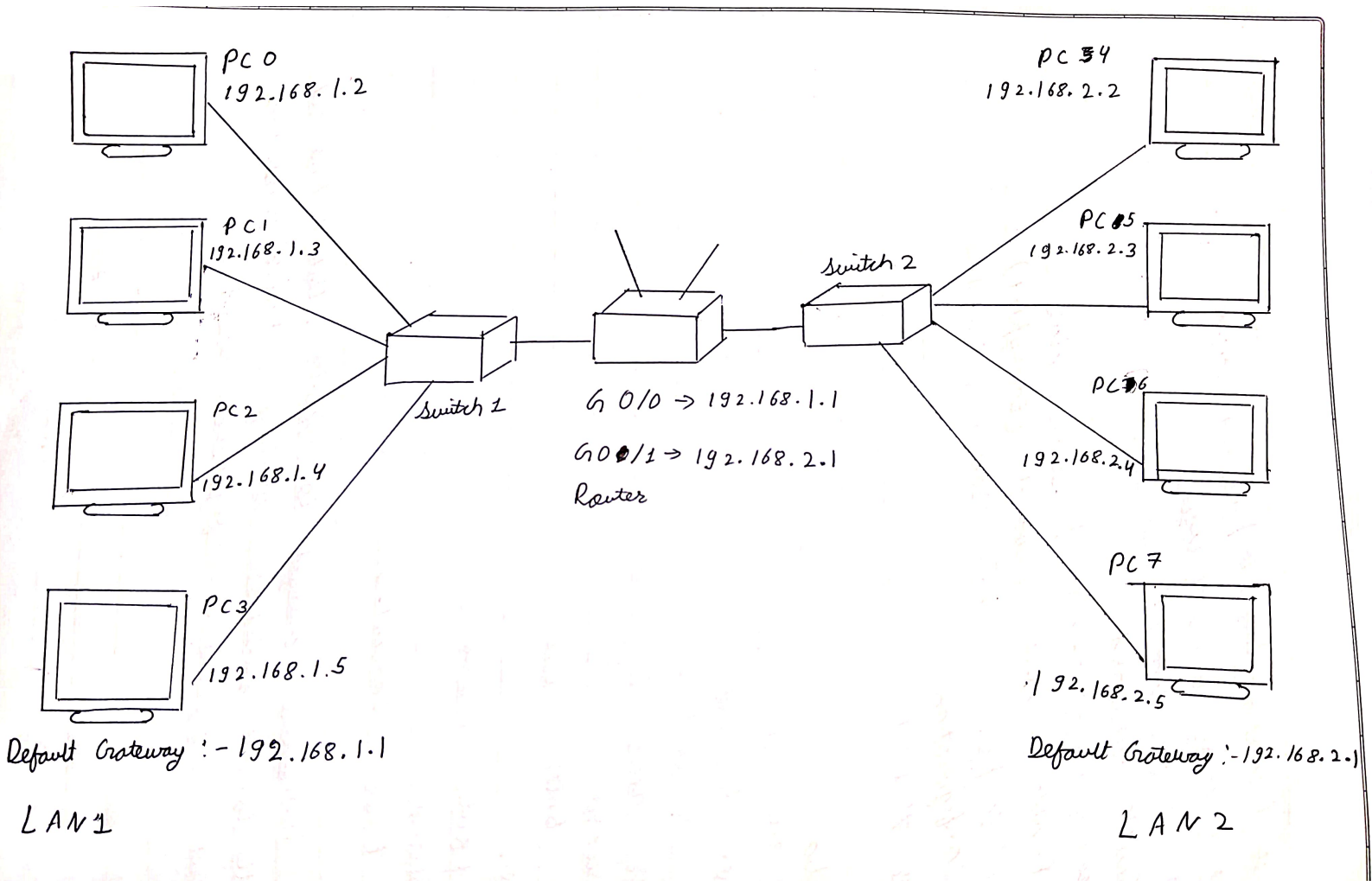
PC 0
192.168.1.2
PC1
192.168.1.3
PC2
192.168.1.4
PC3
192.168.1.5
Switch 1
G 0/0 → 192.168.1.1
G0/1 → 192.168.2.1
Router
Switch 2
PC 4
192.168.2.2
PC5
192.168.2.3
PC6
192.168.2.4
PC7
192.168.2.5
Default Gateway :- 192.168.1.1
LAN1
Default Gateway :- 192.168.2.1
LAN 2

Aim :- To design and simulate a network that connect two separate LANs using a router and enables communication between them.

Requirement :- 1 Router, 2 Switches (2960), 8 PCs (PC0-PC7), Copper straight-through cables

Procedure

- IP Addressing scheme
  - LAN 1 (192.168.1.0)
  - PC0 → 192.168.1.2
  - PC1 → 192.168.1.3
  - PC2 → 192.168.1.4
  - PC3 → 192.168.1.5
  - Default Gateway → 192.168.1.1
  - LAN 2 (192.168.2.0)
  - PC4 → 192.168.2.2
  - PC5 → 192.168.2.3
  - PC6 → 192.168.2.4
  - PC7 → 192.168.2.5
  - Default Gateway → 192.168.2.1

Router Interface

- G0/0 → 192.168.1.1
- G0/1 → 192.168.2.1

1) Create Topology

Connect :

- PC0 - PC3 → Switch 1
- PC4 - PC7 → Switch 2
- Switch 1 → Router G0/0
- Switch 2 → Router G0/1

2) Configure PCs

Go to Desktop → IP Configuration and assign

- IP address
- Subnet mask 255.255.255.0

• Default gateway

3) Configure Router

```
enable
configure terminal
interface g 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
```

4) Verify Configuration

```
show ip interface brief
```

Ensure both interface are up/up

5) Test Connectivity

From PC0:

```
ping 192.168.2.2
```

Result :- The two LAN were successfully connected using a router, and device from one network were able to communicate with device in the other network.